

Github actions

What is GitHub Actions?

GitHub Actions is a CI/CD tool built into GitHub.

It helps you:

-  Automatically build your code
-  Run tests
-  Build Docker images
-  Deploy to AWS / Kubernetes / EKS
-  Run scripts when code changes

What is CI/CD?

- **CI (Continuous Integration)** → Automatically build & test code when pushed.
- **CD (Continuous Deployment/Delivery)** → Automatically deploy application.



1. Event (Trigger)

An **Event** is something that happens in a repository.

Examples:

- push
- pull_request
- workflow_dispatch (manual trigger)
- schedule (cron job)
- issue created

When an event happens, the workflow starts.

👉 Event = Trigger mechanism

2. Workflow

A **Workflow** is a complete automation process.

- Defined in YAML format
- Stored in:
 .github/workflows/
- Can contain multiple jobs

One repository can have multiple workflows.

Example:

- ci.yml
- deploy.yml
- security-scan.yml

👉 Workflow = Full CI/CD pipeline definition

4. Job

A **Job** is a group of steps.

- Runs on a runner
- Can run in parallel with other jobs
- Can depend on other jobs

Example job types:

- Build job
- Test job
- Security scan job
- Deploy job

👉 Job = Logical unit of work

5. Step

A **Step** is a single task inside a job.

Examples:

- Checkout code
- Install dependencies
- Run tests
- Build Docker image

Each step:

- Runs sequentially inside a job
- Can execute a shell command
- Or use a pre-built action

👉 Step = Small execution unit

6. Action

An **Action** is a reusable automation component.

Instead of writing everything from scratch, you can use ready-made actions from GitHub Marketplace.

Types of actions:

- JavaScript actions
- Docker container actions
- Composite actions

Example actions:

- actions/checkout → Downloads repo code
- actions/setup-node → Installs Node.js
- docker/login-action → Logs into DockerHub

👉 Action = Reusable automation building block

7. Runner

A **Runner** is the machine that executes the workflow.

Two types:

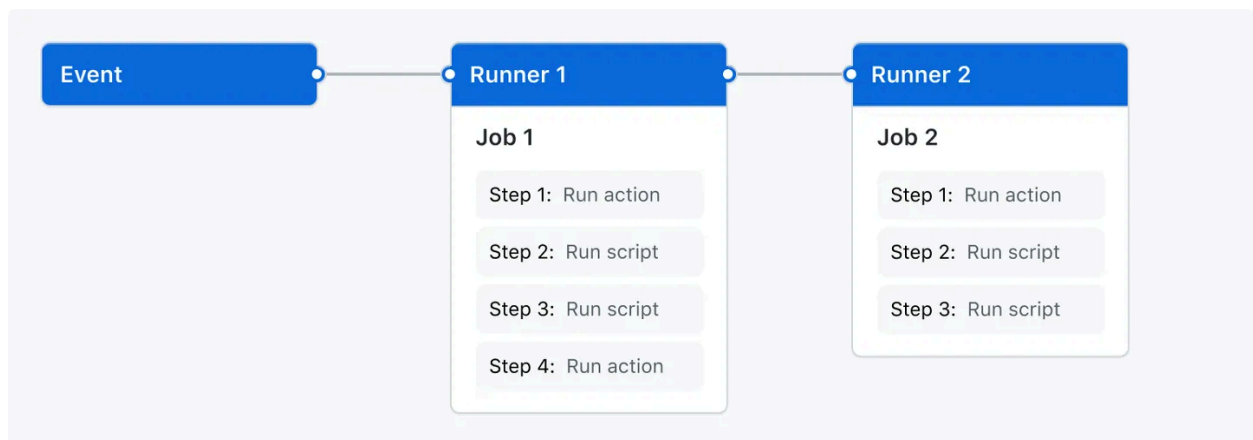
◆ GitHub-hosted Runner

- Managed by GitHub
- Temporary VM
- Automatically created and destroyed
- Supports Ubuntu, Windows, macOS

◆ Self-hosted Runner

- Your own server (EC2, local machine, etc.)
- You manage it
- Useful for custom environments

👉 Runner = Execution environment



When an **event** (like a push or pull request) happens, it triggers a workflow that contains multiple **jobs**. Each job runs on a separate **runner** (a temporary virtual machine provided by GitHub). Inside each runner, the job executes multiple **steps**, and those steps run one after

another in sequence. However, different jobs can run in parallel on different runners, which makes the process faster and more efficient.